

Instruction



Level 1

Grasp a cup

Level 2

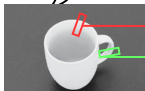
grasp a cup with handle

Level 3

pass me the cup with water

semantic understanding

Grasp a cup **with water** for drinking



grasp the rim



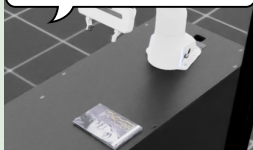
grasp the handle



The trajectory should keep the balance of cup to avoid spilling water

Singulated

Grasp a DVD



Cluttered

Grasp a box from stack



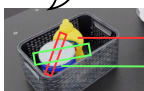
Constrained Space

Grasp a can from basket



scene understanding

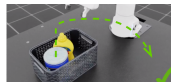
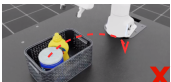
Grasp a can from basket and put on right



limited space



avoid collision



The trajectory should avoid the collision with basket and other objects

Metrics

Detection success rate

Grasp success rate

Task success rate

Efficiency

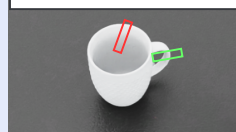
SPL

Execution Time

Task Execution

Grasp a cup **with water**

Detection success



Grasp success



Task success

